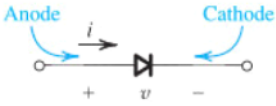
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

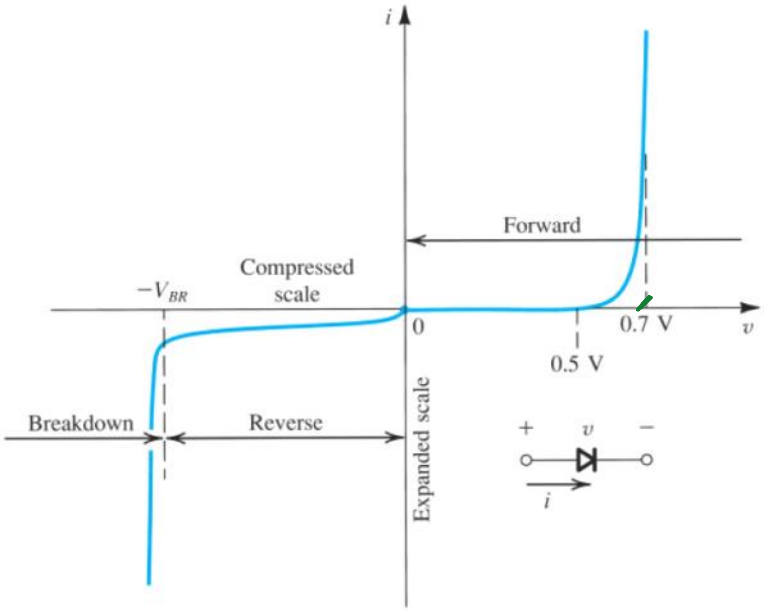
- Chapter 4.4 The Small-Signal Model
- Chapter 4.5 Voltage Regulation
- Chapter 4.6 - Rectifier Circuits
- Chapter 4.7 - Other Diode Applications

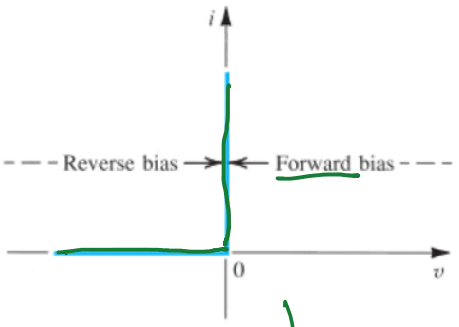
Recap (Diode Model)

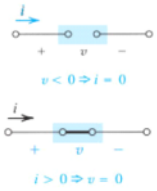


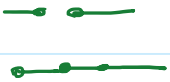
$kT/q \approx 25mV$

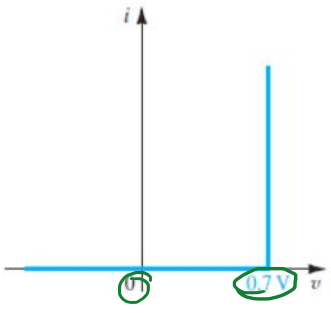
$$i = I_S\left(e^{v/V_T} - 1\right)$$

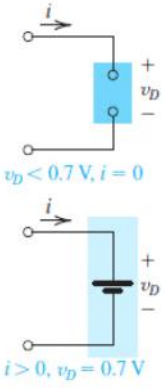


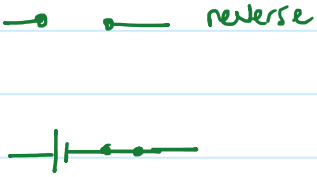




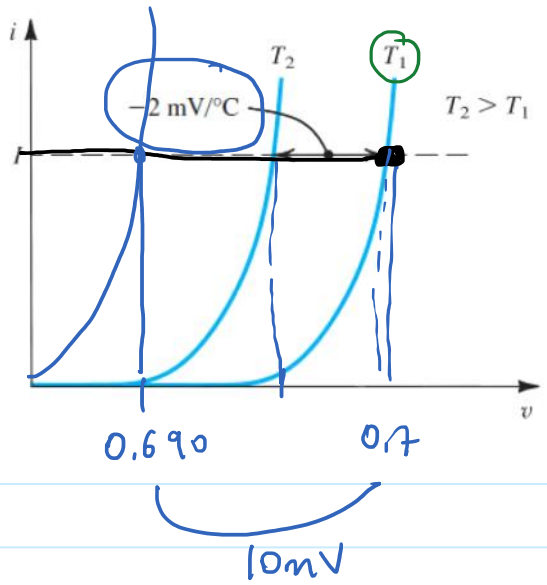






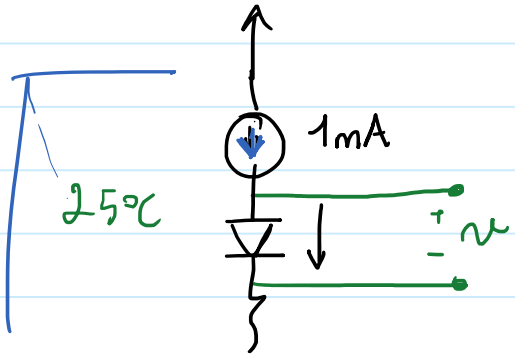


Diode as Temperature Sensor



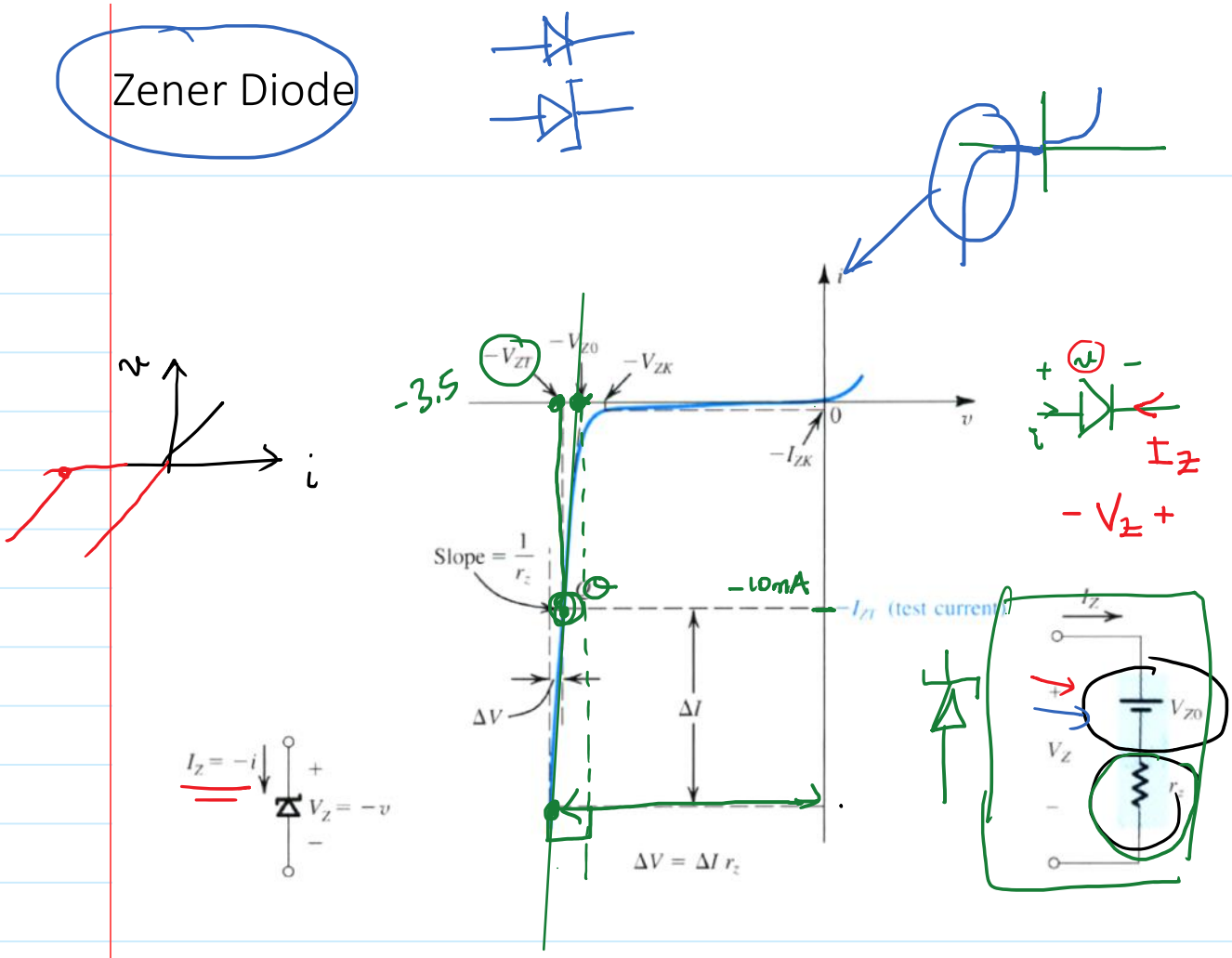
$I_D = I_S \left(e^{V/V_T} - 1\right)$

$V_T = \frac{kT}{q}$



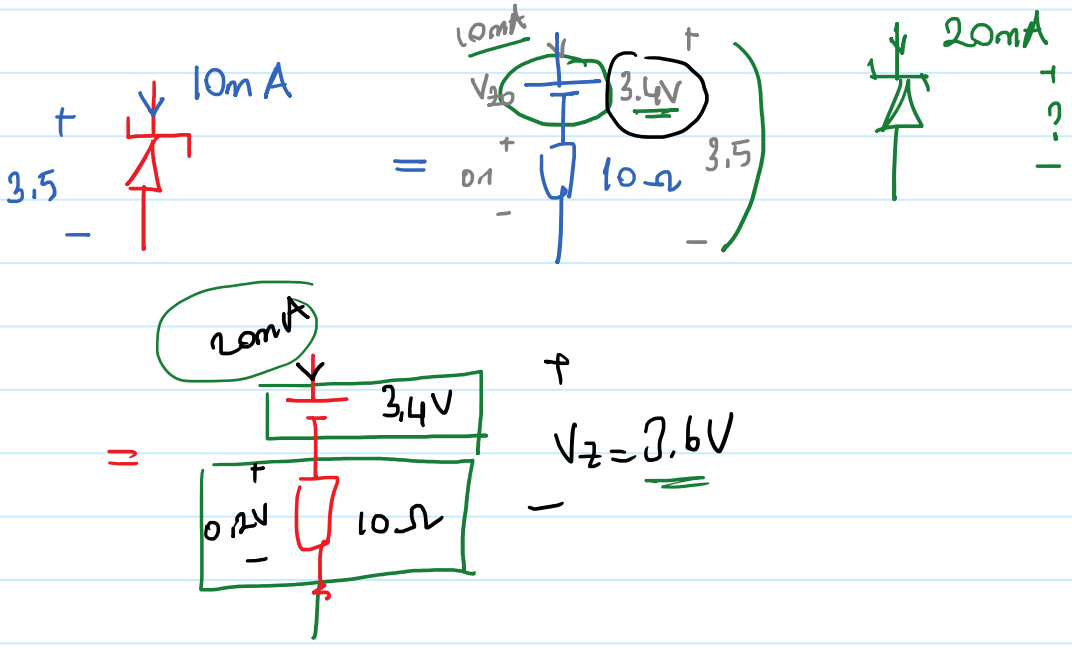
26°C

Zener Diode

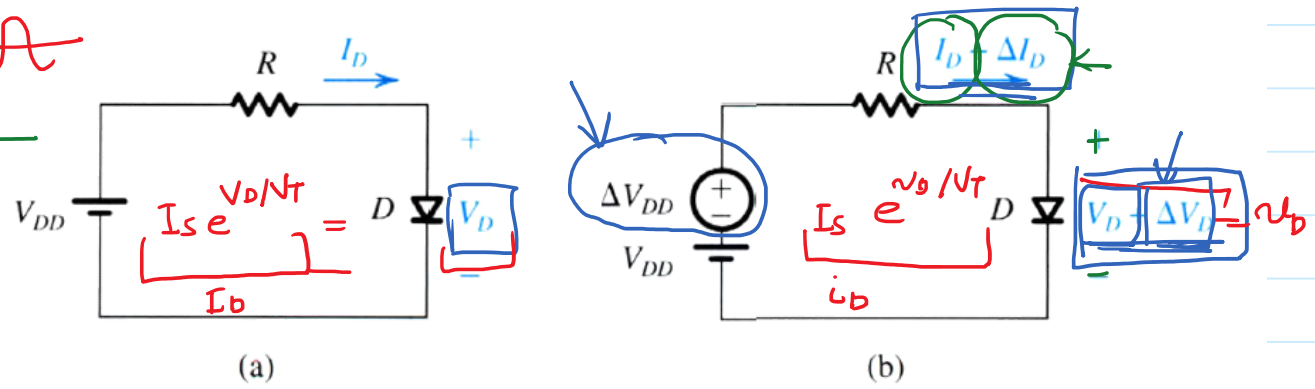


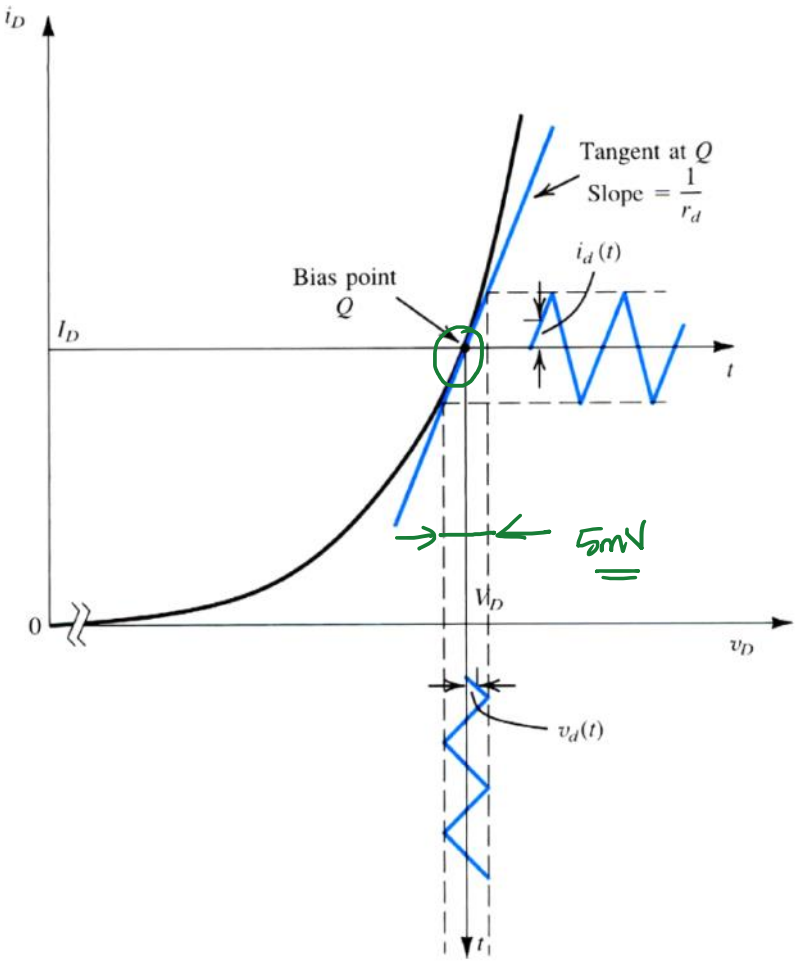
4.16 A 200-mW 3.5-V zener diode is specified at a test current of 10 mA. Its incremental resistance is 10 Ω. What voltage will result when conducting a reverse current of 20 mA? What is the maximum current it can safely dissipate?

Ans. 3.6 V, 51 mA



The Small-Signal Model





The Small-Signal Model (cont.)

$$v_D = V_D + v_d$$

v_D: mixed, V_D: dc, v_d: small signal

$$i_D = I_S e^{v_D/V_T}$$

$$= I_S e^{(V_D + v_d)/V_T}$$

$$= I_S e^{V_D/V_T} \cdot e^{v_d/V_T}$$

$$= I_D \cdot e^{v_d/V_T}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$i_D = I_D \cdot \left(1 + \frac{v_d}{V_T} + \left(\frac{v_d}{V_T}\right)^2 \cdot \frac{1}{2!} + \cdots\right)$$

$V_T = 25\,mV$, $v_d = 5\,mV$

$$i_D \overset{\text{if } x \ll 1}{\approx} I_D \left(1 + \frac{v_d}{V_T}\right) = I_D + \frac{I_D}{V_T} \cdot v_d$$

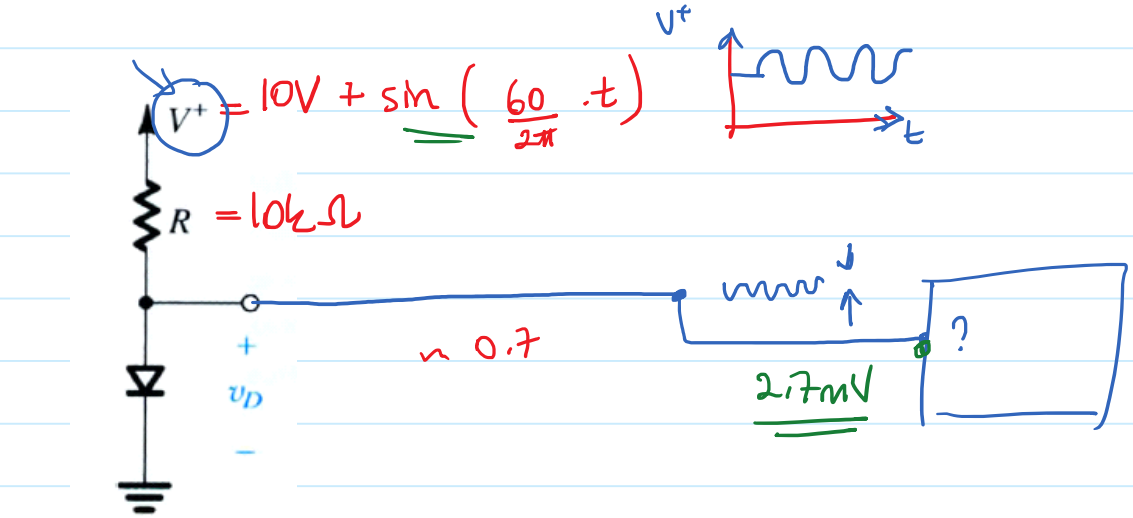
I_D: $\Delta I_D = i_d$

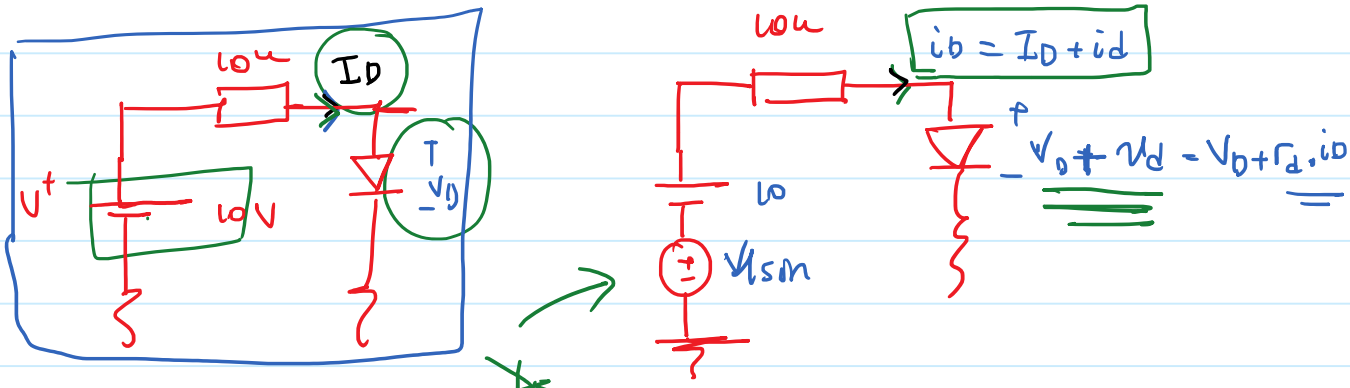
$$i_d = \frac{I_D}{V_T} \cdot v_d \Rightarrow \frac{v_d}{i_d} = \frac{V_T}{I_D} = r_d$$

Example

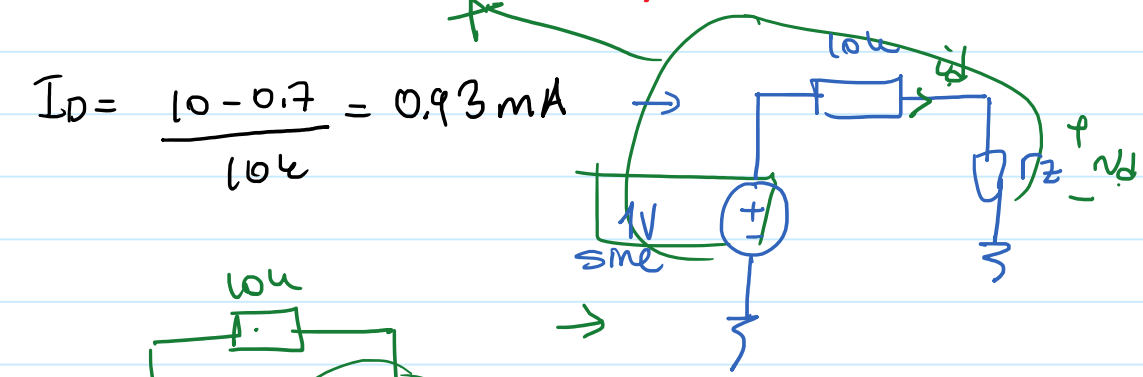
Example 4.5

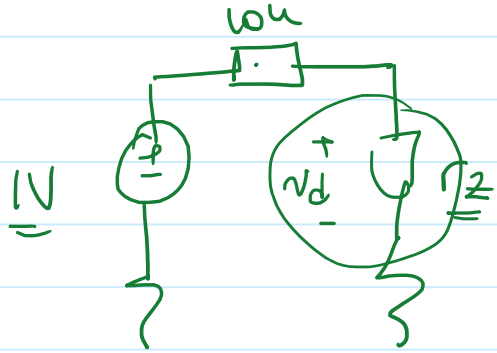
Consider the circuit shown in Fig. 4.17(a) for the case in which $R = 10\,\text{k}\Omega$. The power supply V^+ has a dc value of 10 V on which is superimposed a 60-Hz sinusoid of 1-V peak amplitude. (This "signal" component of the power-supply voltage is an imperfection in the power-supply design. It is known as the **power-supply ripple**. More on this later.) Calculate both the dc voltage of the diode and the amplitude of the sine-wave signal appearing across it. Assume the diode to have a 0.7-V drop at 1-mA current.





$$I_D = \frac{10 - 0.7}{10k} = 0.93\,\text{mA}$$





$$r_z = \frac{V_T}{I_D} = \frac{25m}{0.93m} = 27\Omega$$

$$v_d = 1\,\frac{27}{10k + 27} = 2.7\text{mV} \ll 5\text{mV}$$

Rectifier Circuits

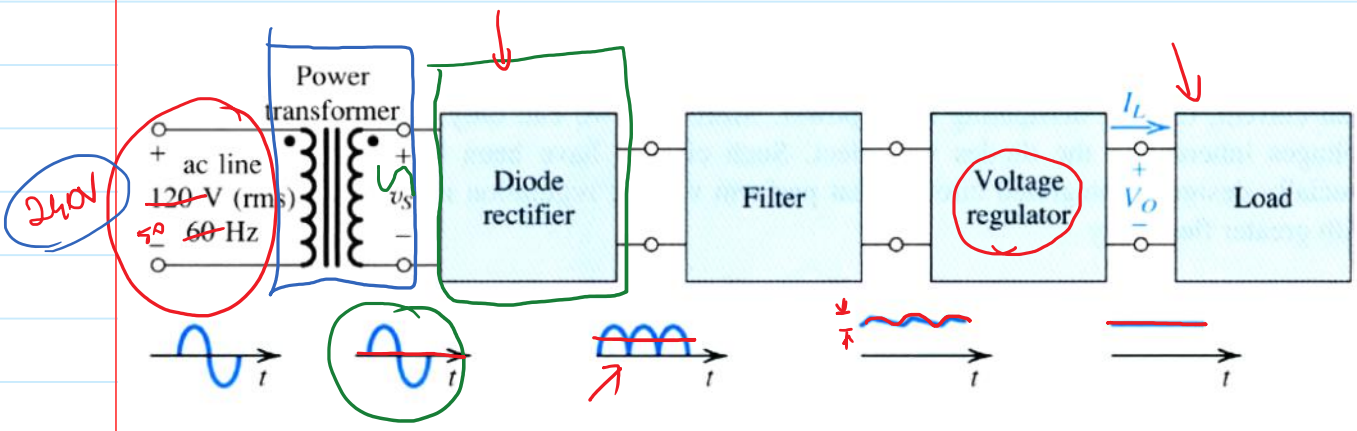
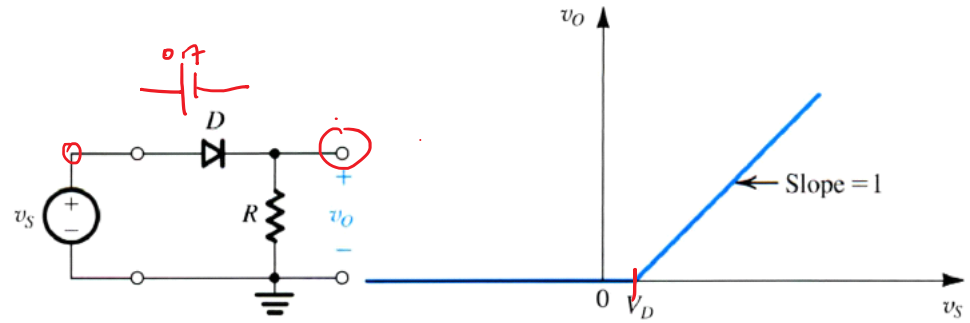
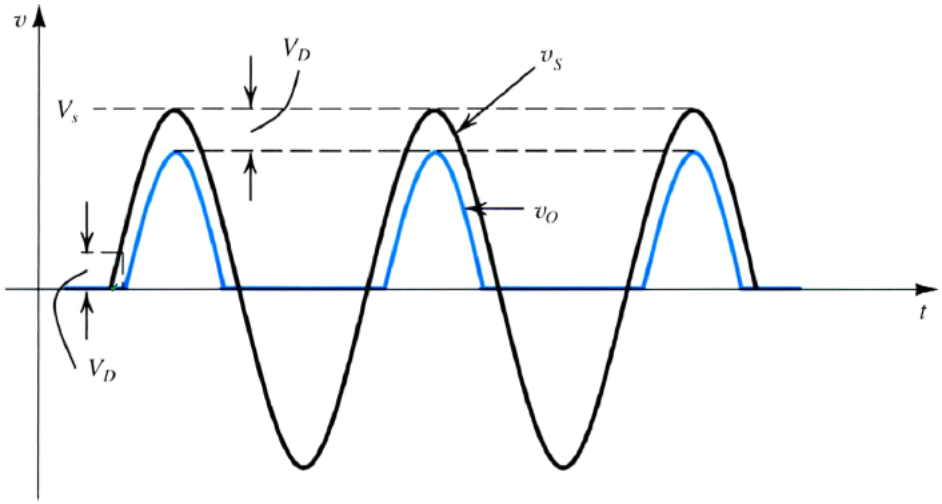


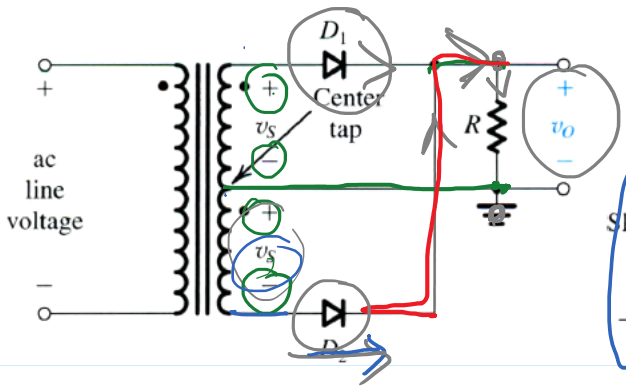
Figure 4.20 Block diagram of a dc power supply.

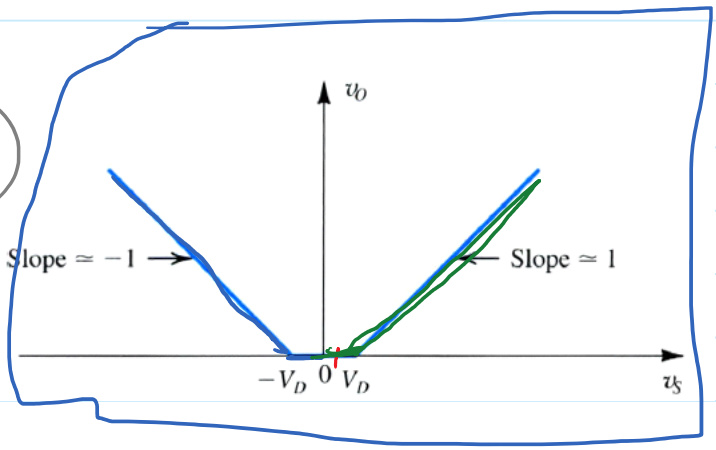
Recap (The Half-Wave Rectifier)

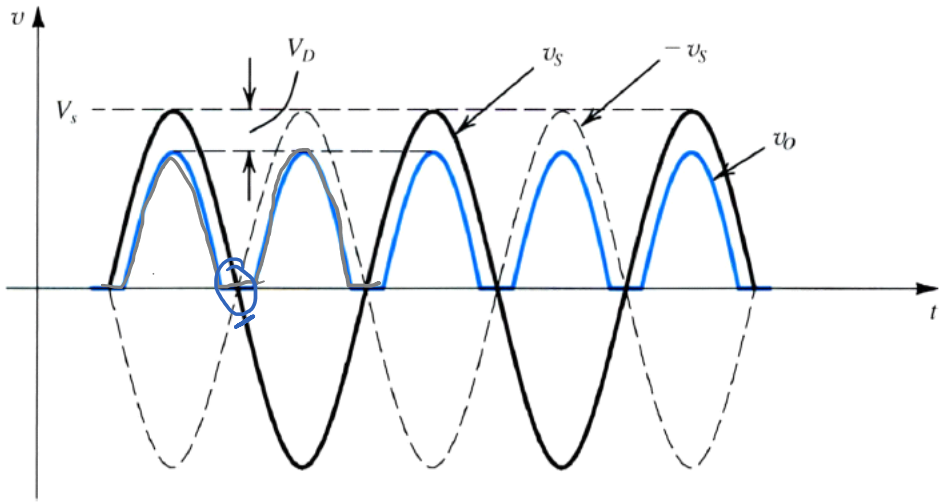


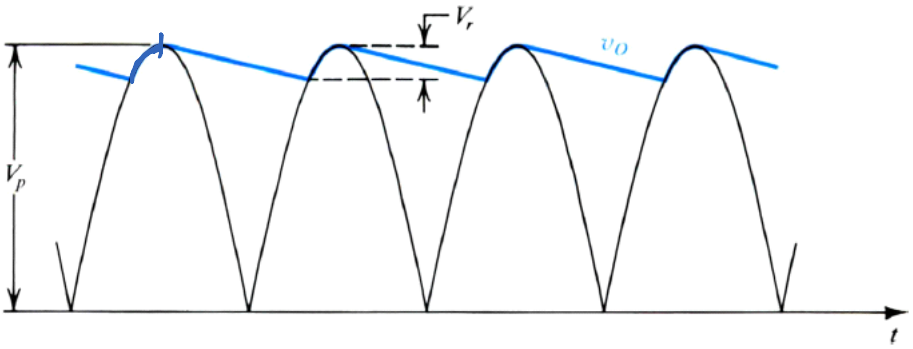


The Full-Wave Rectifier

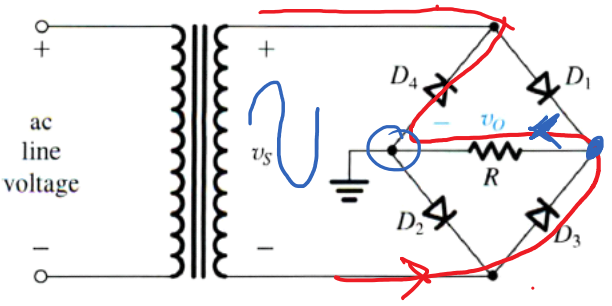


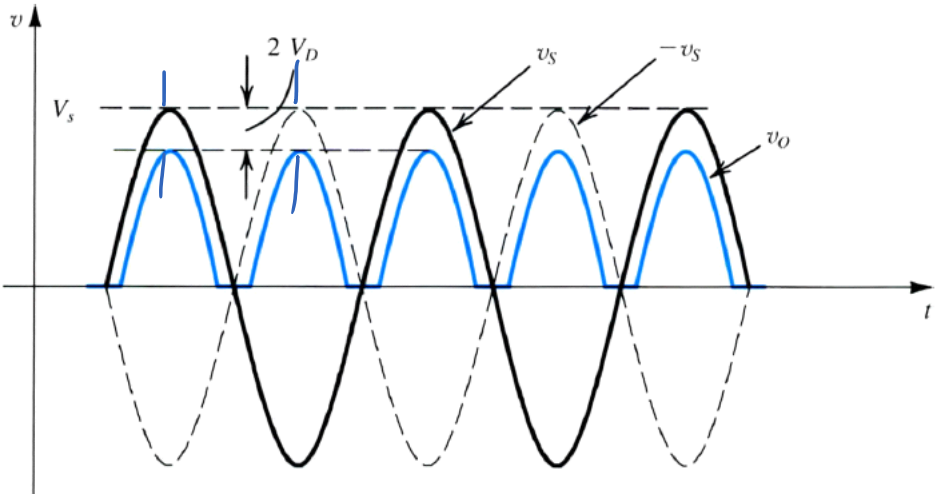




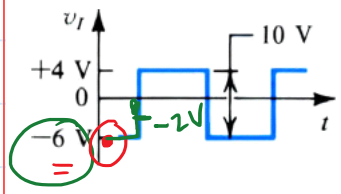


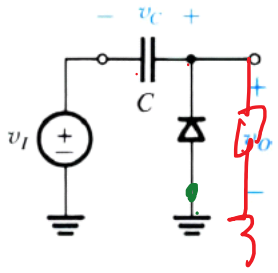
The Bridge Rectifier

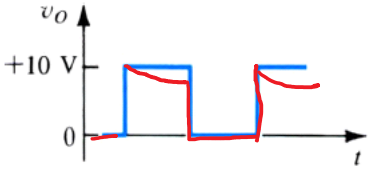


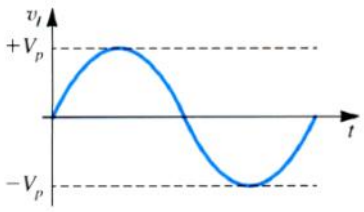


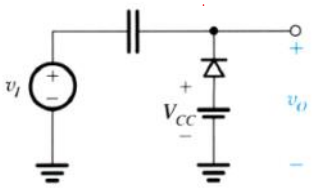
The Clamping Circuit

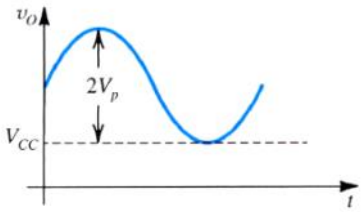


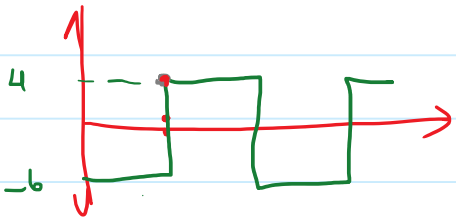


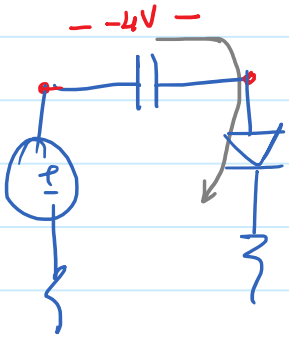


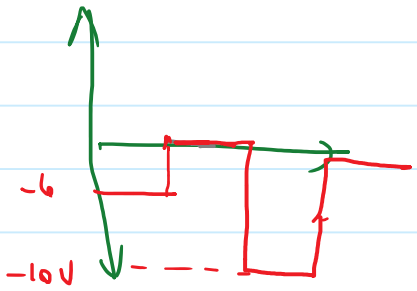




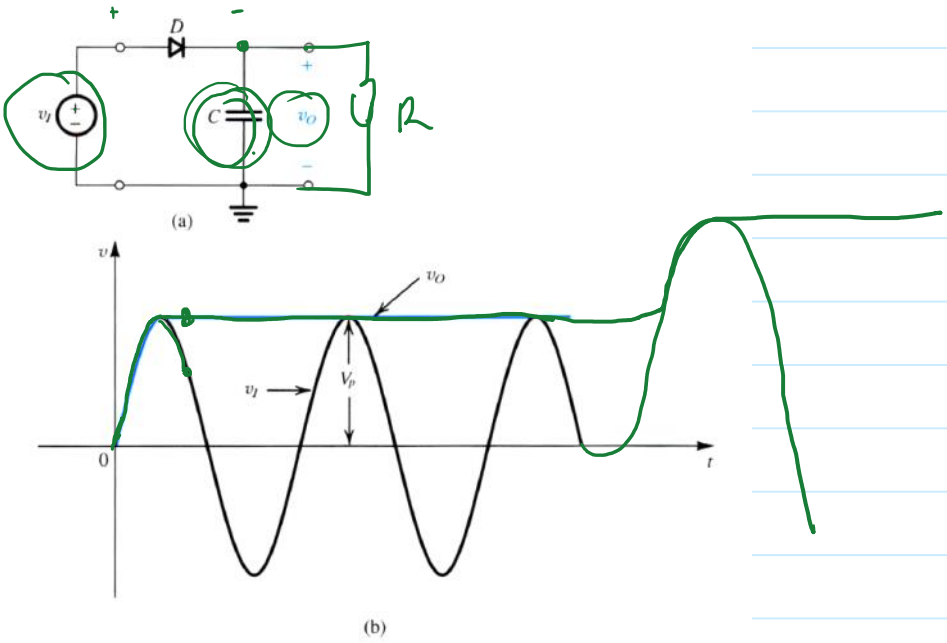


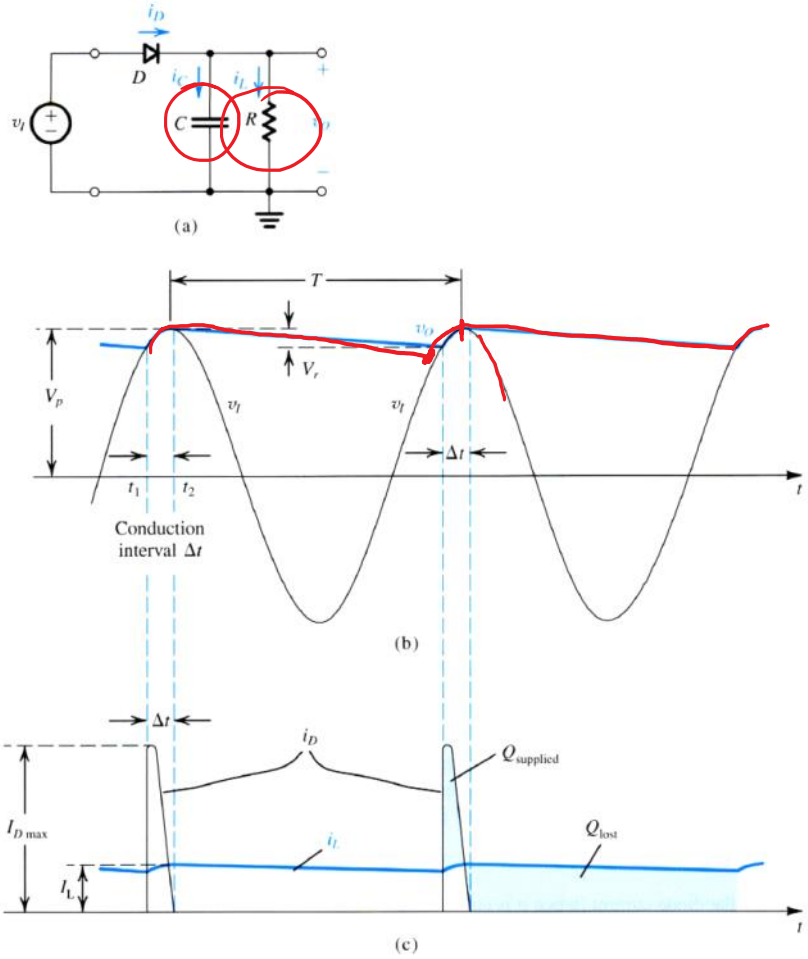




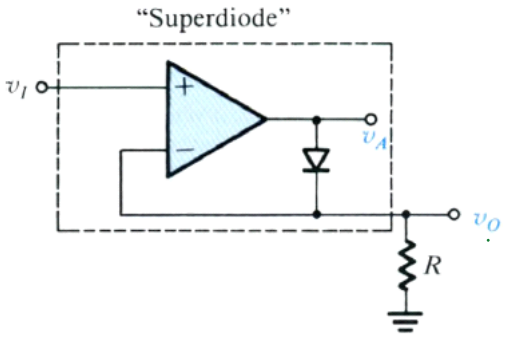


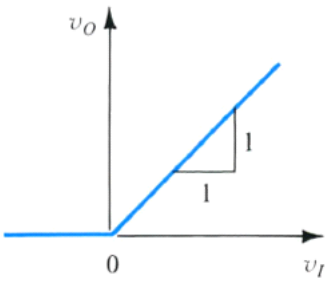
The Peak Rectifier



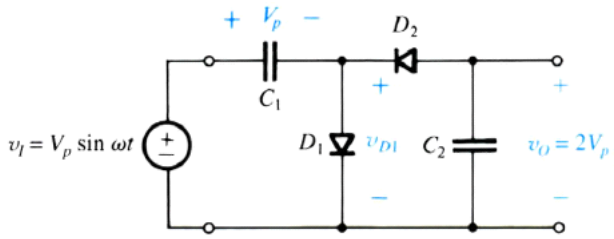


Super-Diode

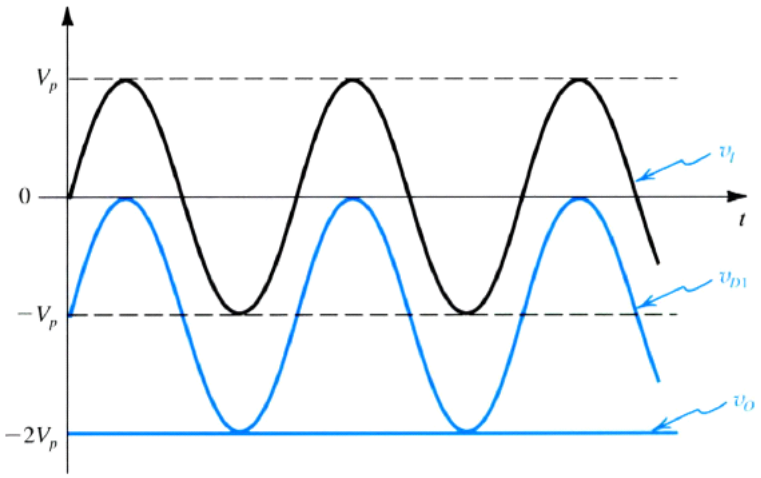




Voltage Doubler



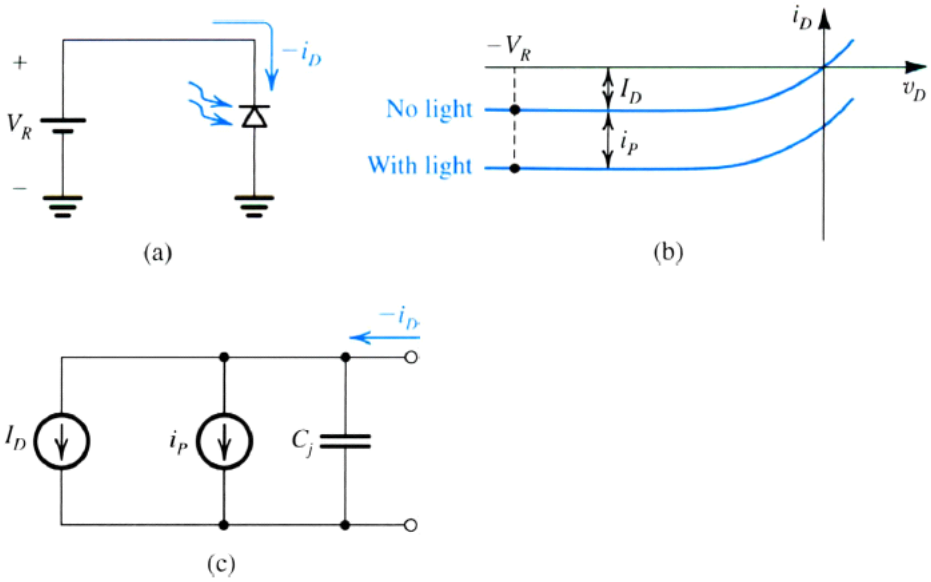
(a)



Varactor

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

Photo-Diode



Light-Emitting Diode (LED)

Template